

DA7066: Programming techniques for Life Sciences

Course introduction

2026-01-19

Who am I?

Kristoffer Sahlin, associate professor in computational biology
(<https://www.su.se/english/profiles/krsa3529-1.475542>)

Work mainly with computational models and algorithms to process sequencing data (mapping, assembly, clustering, etc.).

Questions about the course: ksahlin@math.su.se.

Today

- Purpose and aims with the course
- Practical information: teaching, recitation sessions, labs, course literature.
- Chapter 1 in compendium.

Purpose and aims

DA7066: Programming techniques for Life Sciences

Programming is the process of creating a set of instructions that tell a computer how to perform a task. ([Khan Academy - What is Programming?](#))

The course is about **computer programming**:

- Programming in a modern and popular language (Python).
- Basic:
 - ▶ programming concepts (loops, functions, ..).
 - ▶ data structures in Python (lists, tuples, dictionaries).
- Problem solving by subproblems.
- Program structuring by modularization.
- ...

DA7066: Intended Learning Outcomes

- Independently solve computational problems
- Translate a text description of a problem into code
- Write small programs
- Using programs in a Unix environment.
- Read, understand, modify code.
- Problem solving by subproblems and modularization.
- Apply and explain rules for good programming style

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DA7066 is NOT intended to:

- Be a course about data analysis.
- Teach you commands from some library to do data analysis
 - ▶ pandas
 - ▶ scikit-learn
 - ▶ matplotlib/seaborn
- No Machine Learning.

Algorithms

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Examples:

- A receipt for cooking food
- Notes for playing music
- Sorting numbers or other elements
- Mathematical computations and simulations. (generating primes, cryptography, finite element method (diff eqn), ...)
- ...

Python

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Used for all sorts of applications

Used by the largest IT companies, ex. Instagram, Google, YouTube, Spotify, Amazon...

<https://www.python.org/>
[https://en.wikipedia.org/wiki/Python\(programming_language\)](https://en.wikipedia.org/wiki/Python(programming_language))

Why learn computer programming?

Useful: Software exists everywhere (cellphones, computers, cars, ...)

“Whether you want to uncover the secrets of the universe, or you just want to pursue a career in the 21st century, basic computer programming is an essential skill to learn.” - Stephen Hawking

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Fun and creative

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This course focuses on developing *practical* skills in programming, through computer labs and a project – we *expect you* to work actively full-time throughout the course (40h/week)

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Note: Learning curve is pretty steep if you have never programmed before.

Practical information

Teachers

Lecturer and examiner: Kristoffer Sahlin

Teaching assistants:

- Ivan Tolstoganov (Recitation, labs)
- Sami Aydin (labs)

Course webpage

Course web page (moodle):

<https://kurser.math.su.se/course/view.php?id=1742>

Note: Check that you are registered with an email address that you read frequently.

Examination

Three separate examinations (Ladok):

- LABO (3hp): 3 computer labs (G/U)
- INDU (1.5hp): Project (A-F)
- THEO (3hp): Exam (A-F)

Final grade is a combination of the grade on the exam and the project (exam weighted slightly more).

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All the above are **individual** assignments

Important: read the rules in the beginning of the compendium with the lab assignments.

Course material

A **course compendium** with accompanying exercises have been designed for the course specifically and is freely available on the course web page.

If you find a typo, feel free to report it to me

Computer labs available on the course web page.

A document on guidelines on how to write good code (written by Lars Arvestad) is available on the course web page.

Computer labs

There are 3 in total, available already now.

- ① Hand in through course webpage.
- ② Presented to TAs during the lab hours. You should be able to:
 - ▶ run the code
 - ▶ demonstrate valid output
 - ▶ Present what the code is doing (detailed, since you wrote it)
 - ▶ Answer questions from the TA.

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Deadlines for labs:

- Submission: Mondays 23:59 (Jan 26, Feb 2, Feb 9)
- Presentation: Following Tuesdays during lab hour (present earlier if possible to prevent bottleneck).

Note: All assignments are compared automatically, suspected cheating are reported to the University's disciplinary board!

Computer labs (ctd..)

- Lab 1: Based on chapters/lectures 1, 2, and 3 (functions, conditionals, input, lists)
- Lab 2: Based on chapters/lectures 4 and 6 (file handling, error handling, dictionaries)
- Lab 3: Based on chapters/lectures 7 and 8 (program structure, plotting, arguments and command line)

Exam and project

THEO: Written exam **sometime 16th - 18th of Feb TBD**

- Dont forget to register! (In Ladok, 10 days before)

INDU: Think of it as a larger lab (deadline: 23:59, Feb 18th)

- Submit through moodle
- Random subset of students selected to present for TAs

More info on projects later

Schedule ([link here](#))

- Mondays, Tuesdays, Fridays, 09:00–15:00:
 - ▶ Lectures (9-11)
 - ▶ Recitation session (11-12)
 - ▶ Labs (13-15)
- Wednesdays and Thursdays are for full-time self study:
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- **Lectures:** Lecture 1 -> Chapter 1, Lecture 2 -> Chapter 2, ...
- **Recitation session:** Exercises at the end of each chapter.
- **Labs:**
 - ▶ Help with computer labs/project
 - ▶ Presentations
 - ▶ **Prepare:** by reading through the lab and try solving it **before** the lab hours.

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Exam, labs, and project are separate elements in Ladok, so if you miss a lab, you can still report 4.5 credits.

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- ➌ You can discuss among eachother about programming and various techniques, but not exchange ready or near-ready solutions.
- ➍ If you discover that someone has posted a question on a site about an assignment from our labs, please be kind and inform us. You won't face any consequences for being open about it, and you'll avoid being flagged for plagiarism in our system.

Questions?